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ACCEPTANCE-AWARE DRAFT MODEL TRAINING FOR SPECULATIVE DECODING

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ABSTRACT

Speculative decoding accelerates large language model (LLM) inference by using a lightweight draft model to generate multiple candidate tokens that are subsequently verified by the target model in a single forward pass. The achievable speedup is largely determined by the acceptance length, yet existing draft-model training methods predominantly optimize cross-entropy or Kullback-Leibler (KL) divergence as a proxy for this metric. Although these objectives encourage the draft model to match the target distribution, they do not directly optimize the acceptance length that determines speculative decoding efficiency. More importantly, the acceptance mechanism differs between greedy and sampling-based decoding, making a single training objective suboptimal across decoding regimes.

In this work, we propose acceptance-length-aware training losses that directly optimize the expected number of accepted tokens within a speculative window. We first derive an expected accepted length (EAL) loss that explicitly maximizes the expected accepted length under greedy verification. For sampling-based speculative decoding, we introduce a window total variation (WTV) loss, which directly optimizes the distributional overlap between the temperature-scaled draft and target distributions and accounts for the sequential dependency of acceptance across the speculative window. We further show that both objectives can be combined with a group-relative reinforcement learning stage (GRPO) that uses the simulated acceptance length as a reward, providing an additional gain on top of the supervised objectives.

Extensive experiments across different target models, draft models, tasks, and decoding configurations demonstrate that our acceptance-length-aware losses improve acceptance length over conventional KL-based training. In particular, the WTV loss provides substantial gains under sampling-based decoding, while the EAL loss is better aligned with greedy verification in speculative decoding. Our results demonstrate that training the draft model for the actual acceptance objective, and adapting the objective to the decoding mode, provides a more effective approach to speculative decoding than conventional distribution-matching objectives.

1 INTRODUCTION

Large language models (LLMs) have achieved remarkable performance across a wide range of applications, but their autoregressive generation process remains a major bottleneck for efficient inference. Each generated token typically requires a separate forward pass through the target model. Speculative decoding Leviathan et al. (2023); Hu et al. (2026c) addresses this bottleneck by introducing a lightweight draft model that predicts multiple future tokens, which are then verified by the target model in parallel. When the drafted tokens are accepted, multiple tokens can be generated with the cost of a single target-model verification step. The effectiveness of speculative decoding is therefore strongly dependent on the quality of the draft model. In particular, the acceptance length, i.e., the number of consecutively accepted draft tokens in each verification step, directly determines how many tokens can be generated per target-model invocation. Existing approaches have primarily focused on improving draft-model architectures, including parallel decoding heads, autoregressive speculators, and multi-token prediction modules. In contrast, the training objective for these draft

models has received less attention. Most existing methods train the draft model using conventional cross-entropy or Kullback-Leibler (KL) divergence to match the target model's output distribution.

However, distribution matching is only an indirect proxy for speculative decoding performance. KL divergence and acceptance rate share the same global optimum when the draft model can perfectly reproduce the target distribution. In practice, however, draft models are deliberately much smaller than their target models and therefore have limited capacity. Consequently, the draft model must make a compromise when it cannot exactly reproduce the target distribution. Optimizing KL then asks the draft model to minimize distributional discrepancy, whereas speculative decoding ultimately requires a different goal: producing tokens that can be accepted by the target model. This mismatch becomes especially important when the draft model has insufficient capacity to perfectly match the target. LK Loss Samarin et al. (2026) similarly motivates direct acceptance optimization from this observation, showing that distribution matching losses can lead to suboptimal acceptance-oriented solutions.

More importantly, we observe that acceptance itself is not governed by the same mechanism under different decoding regimes. Under greedy decoding, a draft token is accepted when it matches the target model's top-1 prediction. Therefore, the relevant quantity is the probability that the draft predicts the target's greedy token, and the probability of accepting a longer sequence depends on the product of these per-position probabilities as shown in Section 3.2. In contrast, sampling-based speculative decoding has a fundamentally different acceptance mechanism. When tokens are sampled from temperature-scaled distributions, a drafted token is accepted according to the standard rejection-sampling rule. The resulting per-position acceptance probability is equal to the overlap between the temperature-scaled target and draft distributions as we introduce in Section 3.3.

We propose two acceptance-length-aware training objectives tailored to different decoding regimes. For greedy decoding, we introduce a loss that directly optimizes the expected acceptance length by encouraging consecutive agreement with the target model. For sampling-based decoding, we develop a complementary objective that directly targets the acceptance behavior induced by stochastic sampling. Our approach differs from conventional knowledge distillation in two key aspects. First, we optimize the actual acceptance length rather than using distributional similarity as a proxy. Second, we tailor the training objective to the underlying decoding regime. This distinction is particularly important for sampling-based decoding, where the acceptance behavior differs fundamentally from greedy verification. Beyond these two differentiable surrogates, we also investigate whether the acceptance length can be optimized directly as a non-differentiable reward. We adapt Group-Relative Policy Optimization (GRPO) to the draft-model setting by simulating greedy verification on window-level slices of a single forward pass, using the resulting acceptance length, debiased against a frozen reference draft, as the reward signal. This RL stage is applied on top of a supervised checkpoint and is complementary to the surrogate losses.

We evaluate our objectives across multiple benchmarks and model configurations under both greedy and sampling-based speculative decoding. Our results demonstrate that direct acceptance-length optimization improves draft model quality over conventional KL-based training when the objective is matched to the decoding regime. In summary, our contributions are:

- First, we formulate draft-model training directly around expected acceptance length, replacing distributional matching with the actual objective that determines speculative decoding efficiency.
- Second, we derive a expected accepted length (EAL) loss that directly optimizes consecutive top-1 agreement and a window total variation (WTV) loss that directly optimizes temperature-dependent distributional overlap. Both objectives explicitly model the sequential nature of speculative verification, capturing the multiplicative effect of early-token acceptance on longer accepted sequences.
- We demonstrate the effectiveness of the proposed losses across different models, tasks, and decoding regimes, showing that matching the training objective to the actual acceptance mechanism can consistently improve speculative decoding performance.

2 Background and Related work

2.1 Speculative Decoding

Speculative decoding is a widely adopted technique for accelerating autoregressive language model inference by decoupling token generation into a lightweight draft model and a more capable target model Leviathan et al. (2023); Chen et al. (2023). Given a context, the draft model generates multiple candidate tokens, which are then verified by the target model in parallel. Since rejected tokens are corrected using the target distribution, speculative decoding can preserve the target model's output distribution while reducing the number of expensive autoregressive target-model evaluations. The effectiveness of speculative decoding largely depends on how many drafted tokens can be accepted in each speculation step. We refer to this quantity as the acceptance length, which directly determines the number of target-model tokens generated per verification step and therefore strongly affects the achievable decoding speedup. Consequently, improving the quality of the draft model is closely related to improving its acceptance behavior rather than simply improving its language modeling capability.

Speculative decoding can operate under different decoding regimes. Under greedy decoding, a draft token is accepted when it agrees with the target model's selected token. Under sampling-based decoding, acceptance is determined by the overlap between the target and draft probability distributions through the rejection-sampling procedure. Thus, although both settings share the same draft-then-verify framework, their underlying acceptance mechanisms are fundamentally different.

2.2 Draft Model Architectures

Early speculative decoding methods used a smaller pretrained language model as the draft model Leviathan et al. (2023); Chen et al. (2023). Subsequent work introduced more lightweight and integrated architectures. MEDUSA Cai et al. (2024) uses multiple decoding heads for parallel token prediction, while EAGLE Li et al. (2024) employs an autoregressive draft head over hidden representations, with further improvements in EAGLE-3 Li et al. (2025). DeepSeek-V3 DeepSeek-AI et al. (2024) incorporates Multi-Token Prediction (MTP) Gloeckle et al. (2024) as a native speculative decoding module, while FR-Spec Zhao et al. (2025) reduces draft overhead through frequency-based vocabulary restriction. These works largely focus on how to efficiently generate draft tokens, while how to train draft models for better acceptance remains an important question.

2.3 Training Losses for Draft Models

Most existing draft-model Cai et al. (2024); Li et al. (2024; 2025) training methods formulate draft training as a knowledge distillation problem. Given the target distribution p and draft distribution q, the draft model is trained to approximate the target model, typically using forward KL divergence or its equivalent cross-entropy objective. However, distribution matching is only an indirect objective for speculative decoding. The ultimate goal of a draft model is not to reproduce the target distribution in isolation, but to generate tokens that can be efficiently accepted by the target model.

More recent studies have started to consider objectives that are more closely related to speculative decoding efficiency. DistillSpec Zhou et al. (2023), for example, investigates alternative divergence measures, including reverse KL and total variation (TV) distance. In particular, TV distance has a direct connection to the acceptance probability under sampling-based speculative decoding. AdaSPEC Hu et al. (2025) further addresses the mismatch between conventional distillation objectives and speculative decoding efficiency through selective distillation.

More recently, LK Losses Samarin et al. (2026) explicitly argues that KL divergence should be viewed as a proxy for acceptance optimization, minimizing KL does not necessarily maximize acceptance. LK therefore introduces losses that directly target token acceptance rather than relying solely on distributional similarity.

Our work follows this acceptance-oriented direction but focuses on a different question: how should the training objective be designed when the goal is acceptance length, and when the decoding regime changes from greedy to sampling-based decoding?

2.4 PRELIMINARIES AND NOTATION

We consider speculative decoding with two autoregressive models sharing the same vocabulary: the target model with next-token distribution $p_t(\cdot) \triangleq p(\cdot \mid x_{<t})$, the draft model with next-token distribution $q_t(\cdot) \triangleq q(\cdot \mid x_{<t})$, both conditioned on the same prefix $x_{<t}$. At each position the draft proposes a token, which is then accepted or rejected by the verification criterion described below.

For inference at sampling temperature $T > 0$, the temperature-scaled distributions are

$$\tilde{p}_t(v) \;\propto\; \exp(\log p_t(v)/T)\,, \qquad \tilde{q}_t(v) \;\propto\; \exp(\log q_t(v)/T)\,. \tag{1}$$

In the greedy limit $T \to 0^+$, both $\tilde{p}_t$ and $\tilde{q}_t$ concentrate on the argmax of their respective unscaled distributions.

Two acceptance criteria are commonly used in speculative decoding, and we refer to them as *greedy* and *sampling* verification. Under **greedy verification**, the target commits to its own most likely token $x_t^\star = \arg\max_{v \in \mathcal{V}} p_t(v)$, and the drafted token is admitted only when it coincides with $x_t^\star$. The single-step acceptance probability is therefore the mass the draft places on the target's choice,

$$a_t^{\text{greedy}} \;\triangleq\; q_t(x_t^\star) \;=\; q(x_t^\star \mid x_{<t})\,, \tag{2}$$

and a rejected position is replaced by $x_t^\star$ itself. Under **sampling verification** Leviathan et al. (2023); Chen et al. (2023), the draft instead samples $\hat{y} \sim q_t(\cdot)$ and the token is admitted with probability $\min\{1,\, p_t(\hat{y})/q_t(\hat{y})\}$. Averaging over the draft's own proposal distribution collapses the acceptance probability to the overlap between the two distributions,

$$a_t^{\text{sample}} \;\triangleq\; \mathbb{E}_{\hat{y}\sim\tilde{q}_t}\left[\min\Big(1,\, \frac{\tilde{p}_t(\hat{y})}{\tilde{q}_t(\hat{y})}\Big)\right] \;=\; \sum_{v\in\mathcal{V}} \min\{\tilde{p}_t(v),\, \tilde{q}_t(v)\} \;=\; 1 - \mathrm{TV}(\tilde{p}_t,\, \tilde{q}_t), \tag{3}$$

and a rejected position is resampled from the target-side residual. When the criterion in use is clear from the context, we drop the superscript and write a_t.

3 ACCEPTANCE-AWARE LOSS DESIGN

3.1 SPECULATIVE WINDOW

A speculative window of size γ starting at position t covers positions $t, t+1, \ldots, t+\gamma-1$; we index positions inside the window by the offset $j = 0, 1, \ldots, \gamma-1$. Let $L_t \in \{0, 1, \ldots, \gamma\}$ denote the number of consecutively accepted draft tokens in the window. Verification halts at the first rejection, so the first k tokens are accepted only if each of them is accepted individually. Writing a_t for the per-position acceptance probability under the criterion in use and treating the per-position acceptances as conditionally independent given the prefix,

$$\mathbb{P}(L_t \geq k) \;=\; \prod_{j=0}^{k-1} a_{t+j}, \qquad \mathbb{E}[L_t] \;=\; \sum_{k=1}^{\gamma} \mathbb{P}(L_t \geq k) \;=\; \sum_{k=1}^{\gamma} \prod_{j=0}^{k-1} a_{t+j}, \tag{4}$$

where the second equality is the tail-sum identity for nonnegative integer-valued random variables. Equation 4 is the common template for both objectives: Section 3.2 instantiates it with $a_t = a_t^{\text{greedy}}$ and Section 3.3 with $a_t = a_t^{\text{sample}}$. We write $\mathcal{W}$ for the set of window start positions in a training sequence.

3.2 EXPECTED ACCEPTED LENGTH LOSS

Under greedy verification, a drafted token at position t is accepted if and only if it equals the target's own choice $x_t^\star = \arg\max_{v\in\mathcal{V}} p_t(v)$. Because the training sequences are produced by the frozen target under greedy decoding, $x_t^\star$ coincides with the ground-truth token at that position, so the per-position acceptance probability of Eq. 2 is the probability the draft assigns to the ground-truth continuation. Throughout this section we abbreviate $a_t \triangleq a_t^{\text{greedy}}$.

For a window of size γ starting at position t, greedy verification accepts the first ℓ drafted tokens and (if $\ell < \gamma$) rejects the $(\ell+1)$-th, so the distribution of the accepted length L_t is

$$\mathbb{P}(L_t = 0) \;=\; 1 - a_t, \tag{5}$$

$$\mathbb{P}(L_t = \ell) \;=\; \left(\prod_{j=0}^{\ell-1} a_{t+j}\right)(1 - a_{t+\ell}), \quad \text{for } 1 \le \ell \le \gamma - 1, \tag{6}$$

$$\mathbb{P}(L_t = \gamma) \;=\; \prod_{j=0}^{\gamma-1} a_{t+j}, \tag{7}$$

where the last case has no rejection term because the window is exhausted before a rejection occurs. The expected accepted length is then

$$\mathbb{E}[L_t] \;=\; \sum_{\ell=0}^{\gamma} \ell \cdot \mathbb{P}(L_t = \ell) \;=\; \sum_{\ell=1}^{\gamma-1} \ell \left(\prod_{j=0}^{\ell-1} a_{t+j}\right)(1 - a_{t+\ell}) \;+\; \gamma \prod_{j=0}^{\gamma-1} a_{t+j}, \tag{8}$$

where the $\ell = 0$ term vanishes. Eq. 8 telescopes into the tail-sum form of Eq. 4, giving the more compact expression

$$\mathbb{E}[L_t] \;=\; \sum_{k=1}^{\gamma} \prod_{j=0}^{k-1} a_{t+j}. \tag{9}$$

As an example, for a window of size $\gamma = 3$, Eq. 8 reads

$$\mathbb{E}[L_t] \;=\; 1 \cdot a_t(1 - a_{t+1}) \;+\; 2 \cdot a_t a_{t+1}(1 - a_{t+2}) \;+\; 3 \cdot a_t a_{t+1} a_{t+2}, \tag{10}$$

which equals

$$\mathbb{E}[L_t] \;=\; a_t \;+\; a_t a_{t+1} \;+\; a_t a_{t+1} a_{t+2}. \tag{11}$$

The expected accepted length is a quantity to be *maximized*, so the corresponding training loss is its negation. The per-window EAL loss and the sequence-level objective are

$$\mathcal{L}_{\mathrm{EAL}}^{(t)}(\theta) \;\triangleq\; -\sum_{k=1}^{\gamma} \prod_{j=0}^{k-1} a_{t+j}^{\mathrm{greedy}}, \qquad \mathcal{L}_{\mathrm{EAL}}(\theta) \;=\; \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \mathcal{L}_{\mathrm{EAL}}^{(t)}(\theta), \tag{12}$$

where θ denotes the draft model parameters and the target distributions are frozen.

3.3 Window TV Loss

Under sampling verification, acceptance is not decided by a single token match but by the overlap of the two temperature-scaled distributions. By Eq. 3, the per-position rejection rate at position t equals the total variation distance between them,

$$r_t \;\triangleq\; \mathrm{TV}(\tilde{p}_t,\, \tilde{q}_t), \qquad a_t^{\mathrm{sample}} \;=\; 1 - r_t. \tag{13}$$

Throughout this section we abbreviate $a_t \triangleq a_t^{\mathrm{sample}}$.

Substituting Eq. 13 into the window template of Eq. 4, the expected accepted length of a window of size γ starting at position t is

$$\mathbb{E}[L_t] \;=\; \sum_{k=1}^{\gamma} \prod_{j=0}^{k-1} a_{t+j} \;=\; \sum_{k=1}^{\gamma} \prod_{j=0}^{k-1} \big(1 - \mathrm{TV}(\tilde{p}_{t+j}, \tilde{q}_{t+j})\big). \tag{14}$$

The direct surrogate is the negation of Eq. 14 itself, evaluated with the draft and target models:

$$\boxed{\mathcal{L}_{\mathrm{WTV}}^{(t)}(\theta) \;\triangleq\; -\sum_{k=1}^{\gamma} \prod_{j=0}^{k-1} \Big(1 - \mathrm{TV}(\tilde{p}_{t+j},\, \tilde{q}_{t+j})\Big), \qquad \mathcal{L}_{\mathrm{WTV}}(\theta) \;=\; \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \mathcal{L}_{\mathrm{WTV}}^{(t)}(\theta).} \tag{15}$$

The target distributions $\tilde{p}_{t+j}$ are frozen, and gradients flow only into the draft parameters θ through $\tilde{q}_{t+j}$; the loss is differentiable because TV is a piecewise-linear function of $\tilde{q}$.

Figure 1 illustrates how $\mathcal{L}_{\mathrm{WTV}}$ is computed, reading the diagram from the bottom up. The bottom row is the conditioning context: the draft is run once over the target-generated sequence $x_1, x_2, \ldots$ under teacher forcing, so at every position t the draft and the frozen target see the same prefix $x_{<t}$. The middle row is what each of them produces there, namely the temperature-scaled distributions $\tilde{q}_t$ and $\tilde{p}_t$; the shaded region where the two histograms overlap is exactly the quantity $\sum_v \min\{\tilde{p}_t(v), \tilde{q}_t(v)\}$ that Eq. 3 identifies with the per-position acceptance probability, so the top row of the figure, the sequence of $a_t = 1 - \mathrm{TV}(\tilde{p}_t, \tilde{q}_t)$ values, is obtained for all positions in parallel from a single draft forward pass and a single target forward pass.

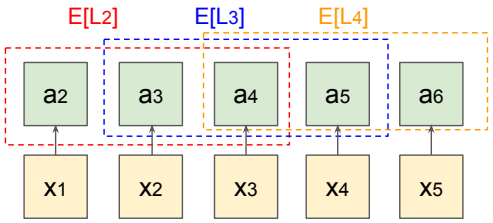


Figure 1: Illustration of the Window TV (WTV) loss with window size $\gamma = 3$. At each position t, the draft $\tilde{q}_t$ and the frozen target $\tilde{p}_t$ are conditioned on the same ground-truth prefix $x_{<t}$, giving $a_t = 1 - \mathrm{TV}(\tilde{p}_t, \tilde{q}_t)$. Each dashed box is a window starting at t, contributing $\mathbb{E}[L_t] = \sum_{k=1}^{\gamma} \prod_{j=0}^{k-1} a_{t+j}$.

The dashed boxes are the speculative windows of Eq. 4. A window starting at t covers positions $t, \ldots, t+\gamma-1$, and because verification halts at the first rejection, the k-th drafted token contributes only through the cumulative product $\prod_{j=0}^{k-1} a_{t+j}$; summing these products over k gives $\mathbb{E}[L_t]$. With $\gamma = 3$ as drawn, the first window contributes $\mathbb{E}[L_t] = a_t + a_t a_{t+1} + a_t a_{t+1} a_{t+2}$. The window then slides by one position and the loss accumulates $-\mathbb{E}[L_t]$ over all start positions.

3.4 GRPO Loss on Acceptance Length

We now describe how we incorporate Group-Relative Policy Optimization (GRPO) Shao et al. (2024); Hu et al. (2026a) into draft model training, using the speculative decoding acceptance length as the reward signal.

3.4.1 Reward construction via simulated greedy verification

For a sequence of length N, we perform three forward passes: (i) the frozen target model produces the per-position tokens $\{x_j^\star\}_{j=1}^N$ of Eq. 2; (ii) the current draft model π_θ produces per-position $\arg\max$ tokens; (iii) a frozen reference draft π_{ref} (a snapshot of π_θ from an earlier training step) produces per-position $\arg\max$ tokens for reward calculation. All three forwards are run once per sequence; individual window-level samples are then obtained by slicing.

For a window of size γ starting at position t, let $z_\theta^{(t)}, z_{\mathrm{ref}}^{(t)} \in \mathbb{R}^{\gamma \times |\mathcal{V}|}$ be the draft logits of the two policies on positions $t, \ldots, t+\gamma-1$. The greedy draft trajectory under each policy is

$$S_k^{\theta,(t)} = \arg\max_{v \in \mathcal{V}} z_{\theta,\,k}^{(t)}(v), \tag{16}$$

$$S_k^{\mathrm{ref},(t)} = \arg\max_{v \in \mathcal{V}} z_{\mathrm{ref},\,k}^{(t)}(v), \qquad k = 1, \ldots, \gamma. \tag{17}$$

Simulating greedy verification against the target, the accepted length of policy π on this window is

$$\tau^{\pi,(t)} = \sum_{k=1}^{\gamma} \prod_{j=1}^{k} \mathbb{1}\left[S_j^{\pi,(t)} = x_{t+j-1}^\star\right], \tag{18}$$

which counts the number of consecutive correct predictions from the start of the window until the first mismatch.

3.4.2 Debiased window reward

Following GTO Hu et al. (2026b), we debias the reward by subtracting the reference policy's acceptance length on the same window:

$$R^{(t)} = \tau^{\pi,(t)} - \tau^{\mathrm{ref},(t)}. \tag{19}$$

Since both policies see the same prefix and are verified against the same target $\arg\max$ sequence, any prefix-dependent difficulty cancels out. The reward $R^{(t)}$ therefore captures only the *incremental*

acceptance length that π_θ achieves over the reference at this window. We stop gradients on $R^{(t)}$: gradients flow only through the policy-ratio term in the surrogate loss, not through the reward itself.

3.4.3 POLICY RATIO ALONG THE CURRENT TRAJECTORY

For the RL objective we also need the log-probability ratio of π_θ to π_{ref} along the trajectory $S^{\theta,(t)}$ actually produced by the current policy. We use a per-token geometric mean to keep the ratio numerically stable across variable acceptance lengths:

$$\log \rho^{(t)} \;=\; \frac{1}{\gamma} \sum_{k=1}^{\gamma} \Big[\log \pi_\theta\big(S_k^{\theta,(t)} \,\big|\, \cdot\big) - \log \pi_{\text{ref}}\big(S_k^{\theta,(t)} \,\big|\, \cdot\big) \Big], \tag{20}$$

where the conditioning $\cdot$ is the ground-truth prefix $x_{1:t+k-1}$. We clip $\log \rho^{(t)}$ to $[-10, 10]$ before exponentiation to avoid overflow.

3.4.4 GROUP SELECTION AND ADVANTAGE NORMALIZATION

To apply GRPO, window-level samples must be partitioned into *groups* within which rewards are standardized. For each input sequence we sample G non-overlapping group anchors uniformly from the valid range. Each group consists of m windows of size γ sliced from the same forward pass. This yields $G \cdot m$ window samples per sequence.

Within each group g, let $\{R^{(t)}\}_{t \in g}$ denote the m debiased rewards. We standardize them to form advantages:

$$A^{(t)} \;=\; \text{clip}\!\left(\frac{R^{(t)} - \bar{R}_g}{\text{std}(R_g) + \varepsilon},\; -C_A,\; C_A \right), \quad \text{(stop gradient)} \tag{21}$$

where $\bar{R}_g$ and $\text{std}(R_g)$ are the group mean and standard deviation, $\varepsilon = 10^{-6}$ avoids division by zero, and C_A caps extreme advantages.

3.4.5 GRPO LOSS

With advantages in hand, we apply a PPO-style clipped surrogate within each group. Let $\rho^{(t)} = \exp(\log \rho^{(t)})$ denote the policy ratio, further clipped to $[\rho_{\min}, \rho_{\max}]$ for safety. The group loss is

$$\mathcal{L}_{\text{GRPO}}^{(g)} \;=\; -\frac{1}{m} \sum_{t \in g} \min\Big(\rho^{(t)} A^{(t)},\;\; \text{clip}\big(\rho^{(t)},\, 1 - \varepsilon_{\text{clip}},\, 1 + \varepsilon_{\text{clip}}\big)\, A^{(t)} \Big), \tag{22}$$

with PPO clipping parameter $\varepsilon_{\text{clip}}$. The sequence-level GRPO loss averages over the G groups:

$$\mathcal{L}_{\text{GRPO}}(\theta) \;=\; \frac{1}{G} \sum_{g=1}^{G} \mathcal{L}_{\text{GRPO}}^{(g)}. \tag{23}$$

4 EXPERIMENT

4.1 SETTINGS

We evaluate our loss designs in a speculative decoding setup with Qwen3 Yang et al. (2025) and Gemma4 Team et al. (2026) model families. We compare the different losses, including $\mathcal{L}_{\text{EAL}}$ in Section 3.2 and $\mathcal{L}_{\text{WTV}}$ in Section 3.3 against several baselines and configurations. The L_{KL} baseline uses only the anchor loss $\mathcal{L} = \mathcal{L}_{\text{anchor}}$, where $\mathcal{L}_{\text{anchor}}$ is the forward KL divergence between the draft and target distributions at each token position. The $L_{KL} + L_{TV}$ baseline adds the per-position TV distance averaged across the window, $\mathcal{L}_{\text{aux}} = \frac{1}{\gamma} \sum_{j=0}^{\gamma-1} \text{TV}(\tilde{p}_{t+j}, \tilde{q}_{t+j})$. The $L_{KL} + L_{EAL}$ baseline adds the differentiable expected-acceptance-length surrogate from Eq. 12 as the auxiliary loss. The L_{LK} baseline is the acceptance-oriented loss of Samarin et al. (2026), which optimizes the per-token acceptance rate instead of the distributional discrepancy. In contrast, the L_{WTV} loss is trained *without* the anchor loss, using only the proposed loss as the full training objective. The temperature T of both $\tilde{p}_t$ and $\tilde{q}_t$ is set to the value used in evaluation. Rows labeled "3 SFT + 3

GRPO" use a two-phase schedule: 3 epochs of SFT with the listed supervised objective, followed by 3 epochs in which the GRPO loss of Section 3.4 is added on top with weight $\lambda_{\text{GRPO}} = 0.1$. The reference policy π_{ref} is the checkpoint at the end of the SFT phase. We apply this schedule on top of both $L_{KL} + 0.1L_{EAL}$ and L_{WTV}, so that the RL stage is evaluated under both acceptance criteria.

The training dataset consists of 60,000 target-generated sequences: prompts are sampled from ShareGPT, and responses are produced by the frozen target model using greedy decoding. Each training sample is the concatenation of prompt and target-generated response, truncated to a maximum length of 2,048 tokens. All training runs are conducted on two NVIDIA H100 GPUs. We use AdamW with a cosine learning-rate schedule, a peak learning rate of $1e^{-5}$ for $L_{KL} + L_{EAL}$ baseline, and gradient clipping. The training hyperparameters are swept for each baseline to get the optimal performance.

We evaluate on three benchmarks covering different task types: MT-Bench, GSM8K, and HumanEval, LongWriter, and STEM QA. Specifically, we use a tree of maximum depth $\gamma = 7$ with per-depth top-K branching factors $(k_1, k_2, k_3, k_4, k_5, k_6, k_7) = (4, 3, 2, 1, 1, 1, 1)$. The fully-expanded tree is then dynamically pruned at runtime to a total token budget of 128 nodes by retaining the nodes with the highest cumulative draft scores along their root-to-node paths. We also evaluate the chain speculative decoding with $\gamma = 7$. Evaluations are conducted on one H100 GPU with greedy and rejection sampling Li et al. (2026) verification, batch size 1, and identical sampling configurations across loss variants to ensure fair comparison. For each benchmark we report acceptance length: the average number of tokens accepted per verification step.

Table 1: Acceptance length on five benchmarks with Qwen3-8B as the target model and Qwen3-0.6B as the draft model with tree-based speculative decoding under greedy verification and sampling temperatures $T = 0.5$ and $T = 1.0$. Colored numbers indicate ranking by Avg. Acceptance Length within each setting: red denotes best, blue denotes second-best, orange denotes third-best.

Setting	Qwen3-8B	Qwen3-0.6B	Acceptance Length						Stat.
	Stage	Loss	MT-Bench	GSM8K	HumanEval	LongWriter	STEM QA	Avg	Med./P25/P75
Greedy	6 SFT	L_{KL}	4.204	5.289	4.896	3.788	4.295	4.494 (0.006)	5/3/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.217	5.299	4.904	3.806	4.290	4.503 (0.007)	5/3/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	4.213	5.322	4.917	3.817	4.303	4.514 (0.005)	5/3/7
	6 SFT	L_{CE}	4.180	5.309	4.885	3.749	4.286	4.482 (0.003)	5/3/7
	6 SFT	L_{LK}	4.213	5.281	4.908	3.805	4.324	4.506 (0.009)	5/3/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	4.198	5.297	4.898	3.802	4.291	4.497 (0.006)	5/3/7
	6 SFT	L_{WTV}	4.148	5.263	4.797	3.688	4.243	4.428 (0.002)	5/3/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	4.169	5.341	4.878	3.724	4.267	4.476 (0.001)	5/3/7
$T = 0.5$	6 SFT	L_{KL}	4.011	5.284	4.510	3.700	4.014	4.304 (0.009)	5/3/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.021	5.290	4.536	3.719	3.998	4.313 (0.005)	5/3/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	4.044	5.295	4.551	3.730	4.009	4.326 (0.005)	5/3/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	4.010	5.278	4.522	3.707	4.004	4.304 (0.006)	5/3/7
	6 SFT	L_{WTV}	4.472	5.671	5.012	4.231	4.512	4.780 (0.018)	6/3/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	4.506	5.722	5.005	4.244	4.504	4.796 (0.010)	6/3/7
$T = 1.0$	6 SFT	L_{KL}	4.805	5.794	4.913	4.765	4.712	4.998 (0.018)	6/3/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.829	5.750	4.951	4.825	4.770	5.025 (0.011)	6/3/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	4.802	5.849	4.903	4.829	4.778	5.032 (0.003)	6/3/7
	6 SFT	L_{CE}	4.805	5.882	5.038	4.795	4.816	5.067 (0.010)	6/4/7
	6 SFT	L_{LK}	4.808	5.798	4.918	4.731	4.713	4.994 (0.012)	6/3/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	4.742	5.800	4.940	4.748	4.621	4.970 (0.004)	6/4/7
	6 SFT	L_{WTV}	5.580	6.436	5.687	5.629	5.416	5.750 (0.012)	7/5/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	5.596	6.453	5.697	5.599	5.472	5.763 (0.019)	7/5/7

4.2 RESULTS

Table 1 reports the acceptance length for Qwen3-8B/Qwen3-0.6B under tree drafting. Under greedy verification the acceptance-oriented objectives cluster tightly: $L_{KL} + 0.1L_{EAL}$ reaches 4.503 against 4.494 for L_{KL}, the subsequent GRPO stage gives the best result at 4.514, and L_{LK} is competitive at 4.506. L_{WTV} is the weakest objective here at 4.428, consistent with the fact that its acceptance model does not match greedy verification. The picture reverses under sampling verification. At $T = 0.5$, L_{WTV} reaches 4.780 against 4.304 for L_{KL} and 4.313 for $L_{KL} + 0.1L_{EAL}$, and at $T = 1.0$ it reaches 5.750 against 4.998; adding GRPO on top of WTV improves these to 4.796 and 5.763.

Table 2 reports the same pair under chain drafting. Under greedy verification the spread between objectives is small: $L_{KL} + 0.1L_{EAL}$ and L_{WTV} both reach 3.064 against 3.051 for L_{KL}, and the GRPO stage on top of WTV gives the best result at 3.065. Under sampling verification the gap widens: L_{WTV} reaches 3.070 at $T = 0.5$ and 3.169 at $T = 1.0$, against 2.922 and 2.823 for L_{KL}, with GRPO adding a further gain to 3.085 and 3.203.

Table 2: Acceptance length on five benchmarks with Qwen3-8B as the target model and Qwen3-0.6B as the draft model with chain-based speculative decoding under greedy verification and sampling temperatures $T = 0.5$ and $T = 1.0$. Colored numbers indicate ranking by Avg. Acceptance Length within each setting: red denotes best, blue denotes second-best, orange denotes third-best.

Setting	Qwen3-8B	Qwen3-0.6B	Acceptance Length						Stat.
	Stage	**Loss**	**MT-Bench**	**GSM8K**	**HumanEval**	**LongWriter**	**STEM QA**	**Avg**	**Med./P25/P75**
Greedy	6 SFT	L_{KL}	2.670	3.826	3.268	2.573	2.919	3.051 (0.002)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	2.682	3.837	3.272	2.618	2.912	3.064 (0.005)	3/1/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	2.679	3.849	3.279	2.597	2.900	3.061 (0.012)	3/1/7
	6 SFT	L_{CE}	2.685	3.815	3.228	2.569	2.926	3.045 (0.003)	2/1/6
	6 SFT	L_{LK}	2.675	3.831	3.270	2.594	2.922	3.058 (0.004)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	2.676	3.833	3.273	2.581	2.945	3.062 (0.004)	3/1/7
	6 SFT	L_{WTV}	2.712	3.856	3.215	2.571	2.965	3.064 (0.003)	3/1/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	2.715	3.861	3.220	2.578	2.952	3.065 (0.008)	3/1/7
$T = 0.5$	6 SFT	L_{KL}	2.614	3.811	2.943	2.515	2.726	2.922 (0.011)	2/1/6
	6 SFT	$L_{KL} + 0.1L_{EAL}$	2.645	3.799	2.978	2.515	2.732	2.934 (0.014)	2/1/6
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	2.656	3.844	3.031	2.545	2.786	2.972 (0.022)	2/1/6
	6 SFT	L_{CE}	2.651	3.846	3.042	2.574	2.776	2.978 (0.019)	2/1/6
	6 SFT	$L_{KL} + 0.1L_{TV}$	2.606	3.779	3.009	2.512	2.736	2.928 (0.009)	2/1/6
	6 SFT	L_{WTV}	2.712	3.929	3.218	2.636	2.853	3.070 (0.024)	3/1/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	2.770	3.883	3.221	2.651	2.899	3.085 (0.021)	3/1/7
$T = 1.0$	6 SFT	L_{KL}	2.407	3.698	2.658	2.689	2.665	2.823 (0.011)	2/0/5
	6 SFT	$L_{KL} + 0.1L_{EAL}$	2.459	3.692	2.681	2.710	2.678	2.844 (0.024)	2/0/5
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	2.424	3.797	2.732	2.755	2.715	2.885 (0.014)	2/0/5
	6 SFT	L_{CE}	2.472	3.768	2.745	2.739	2.707	2.886 (0.021)	2/0/5
	6 SFT	L_{LK}	2.431	3.695	2.667	2.681	2.673	2.829 (0.016)	2/0/5
	6 SFT	$L_{KL} + 0.1L_{TV}$	2.406	3.660	2.702	2.712	2.675	2.831 (0.015)	2/0/5
	6 SFT	L_{WTV}	2.692	3.960	3.190	2.986	3.017	3.169 (0.020)	3/1/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	2.707	3.989	3.223	3.055	3.043	3.203 (0.021)	3/1/7

Table 3: Acceptance length on five benchmarks with Qwen3-8B as the target model and Qwen3-1.7B as the draft model with tree-based speculative decoding under greedy verification and sampling temperatures $T = 0.5$ and $T = 1.0$. Colored numbers indicate ranking by Avg. Acceptance Length within each setting: red denotes best, blue denotes second-best, orange denotes third-best.

Setting	Qwen3-8B	Qwen3-1.7B	Acceptance Length						Stat.
	Stage	**Loss**	**MT-Bench**	**GSM8K**	**HumanEval**	**LongWriter**	**STEM QA**	**Avg**	**Med./P25/P75**
Greedy	6 SFT	L_{KL}	4.840	5.705	5.372	4.407	4.915	5.048 (0.001)	6/4/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.872	5.700	5.375	4.425	4.914	5.057 (0.004)	6/4/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	4.888	5.741	5.410	4.391	4.978	5.082 (0.009)	7/4/7
	6 SFT	L_{CE}	4.803	5.648	5.359	4.335	4.882	5.005 (0.004)	6/3/7
	6 SFT	L_{LK}	4.852	5.703	5.373	4.341	4.919	5.038 (0.006)	6/4/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	4.834	5.726	5.372	4.341	4.933	5.041 (0.008)	6/4/7
	6 SFT	L_{WTV}	4.803	5.726	5.379	4.350	4.957	5.043 (0.005)	6/4/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	4.828	5.727	5.385	4.340	4.920	5.040 (0.016)	6/4/7
$T = 0.5$	6 SFT	L_{KL}	4.793	5.728	5.225	4.552	4.723	5.004 (0.010)	6/4/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.793	5.746	5.256	4.538	4.710	5.009 (0.009)	6/4/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	4.856	5.729	5.254	4.552	4.732	5.025 (0.017)	6/4/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	4.818	5.727	5.221	4.533	4.700	5.000 (0.008)	6/4/7
	6 SFT	L_{WTV}	5.200	6.055	5.602	4.999	5.240	5.419 (0.012)	7/4/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	5.218	6.069	5.634	5.005	5.233	5.432 (0.007)	7/5/7
$T = 1.0$	6 SFT	L_{KL}	5.644	6.265	5.673	5.678	5.385	5.729 (0.019)	7/5/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	5.678	6.309	5.670	5.635	5.359	5.730 (0.013)	7/5/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	5.615	6.304	5.768	5.667	5.419	5.755 (0.016)	7/5/7
	6 SFT	L_{CE}	5.982	6.539	6.213	6.049	5.804	6.117 (0.014)	7/6/7
	6 SFT	L_{LK}	5.662	6.284	5.671	5.696	5.504	5.763 (0.015)	7/5/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	5.661	6.293	5.695	5.654	5.373	5.735 (0.009)	7/5/7
	6 SFT	L_{WTV}	6.272	6.775	6.419	6.350	6.256	6.414 (0.023)	7/7/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	6.379	6.764	6.382	6.413	6.345	6.457 (0.018)	7/7/7

Table 3 reports the acceptance length for Qwen3-8B/Qwen3-1.7B under tree drafting. With this larger draft model the greedy column compresses further: L_{KL} itself reaches 5.048, $L_{KL} + 0.1L_{EAL}$ 5.057, and the GRPO stage 5.082, while the remaining objectives all fall within 0.05 of the baseline. The advantage of WTV again appears only under sampling verification, where it reaches 5.419 at $T = 0.5$ and 6.414 at $T = 1.0$ against 5.004 and 5.729 for L_{KL}, with GRPO improving these to 5.432 and 6.457.

Table 4 reports the same pair under chain drafting. Under greedy verification the GRPO stage on top of $L_{KL} + 0.1L_{EAL}$ gives the best result at 3.698, ahead of L_{WTV} at 3.659 and $L_{KL} + 0.1L_{EAL}$ at 3.658; L_{CE} is a clear outlier at 3.546. Under sampling verification L_{WTV} leads at both temperatures, reaching 3.671 at $T = 0.5$ and 3.767 at $T = 1.0$ against 3.624 and 3.593 for L_{KL}, and GRPO improves these to 3.676 and 3.789.

Table 5 reports the acceptance length for Qwen3-32B/Qwen3-4B under tree drafting. Under greedy verification, $L_{KL} + 0.1L_{EAL}$ reaches 5.073 against 5.057 for L_{KL} and the GRPO stage gives the best result at 5.088, with L_{LK} third at 5.063; L_{WTV} is last at 5.039. Under sampling verification

Table 4: Acceptance length on five benchmarks with Qwen3-8B as the target model and Qwen3-1.7B as the draft model with chain-based speculative decoding under greedy verification and sampling temperatures $T = 0.5$ and $T = 1.0$. Colored numbers indicate ranking by Avg. Acceptance Length within each setting: red denotes best, blue denotes second-best, orange denotes third-best.

Setting	Qwen3-8B	Qwen3-1.7B	Acceptance Length						Stat.
	Stage	Loss	MT-Bench	GSM8K	HumanEval	LongWriter	STEM QA	Avg	Med./P25/P75
Greedy	6 SFT	L_{KL}	3.265	4.399	3.859	3.183	3.537	3.649 (0.001)	4/1/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	3.274	4.413	3.871	3.180	3.551	3.658 (0.004)	4/1/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	3.320	4.431	3.892	3.250	3.598	3.698 (0.015)	4/1/7
	6 SFT	L_{CE}	3.108	4.295	3.794	3.074	3.458	3.546 (0.006)	3/1/7
	6 SFT	L_{LK}	3.270	4.405	3.864	3.202	3.535	3.655 (0.003)	4/1/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	3.260	4.399	3.871	3.188	3.550	3.654 (0.004)	4/1/7
	6 SFT	L_{WTV}	3.267	4.396	3.858	3.179	3.593	3.659 (0.002)	4/1/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	3.295	4.393	3.869	3.155	3.564	3.655 (0.013)	4/1/7
$T = 0.5$	6 SFT	L_{KL}	3.273	4.465	3.671	3.299	3.413	3.624 (0.005)	4/1/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	3.299	4.461	3.671	3.306	3.458	3.639 (0.004)	4/1/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	3.317	4.464	3.700	3.293	3.427	3.640 (0.013)	4/1/7
	6 SFT	L_{CE}	3.154	4.278	3.702	3.227	3.302	3.533 (0.027)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	3.294	4.388	3.726	3.261	3.447	3.623 (0.005)	4/1/7
	6 SFT	L_{WTV}	3.312	4.481	3.774	3.311	3.475	3.671 (0.018)	4/1/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	3.348	4.480	3.807	3.311	3.433	3.676 (0.010)	4/1/7
$T = 1.0$	6 SFT	L_{KL}	3.122	4.340	3.554	3.554	3.394	3.593 (0.014)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	3.132	4.389	3.557	3.564	3.337	3.596 (0.017)	3/1/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	3.140	4.389	3.583	3.568	3.407	3.617 (0.011)	3/1/7
	6 SFT	L_{CE}	3.173	4.372	3.748	3.604	3.436	3.667 (0.036)	3/1/7
	6 SFT	L_{LK}	3.127	4.362	3.555	3.544	3.410	3.600 (0.015)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	3.125	4.352	3.500	3.529	3.359	3.573 (0.016)	3/1/7
	6 SFT	L_{WTV}	3.252	4.553	3.787	3.707	3.537	3.767 (0.029)	4/1/7
	3 SFT + 3 GRPO	$L_{WTV} \rightarrow L_{WTV} + 0.1L_{GRPO}$	3.299	4.549	3.792	3.757	3.549	3.789 (0.009)	4/1/7

Table 5: Acceptance length on five benchmarks with Qwen3-32B as the target model and Qwen3-4B as the draft model with tree-based speculative decoding under greedy verification and sampling temperatures $T = 0.5$ and $T = 1.0$. Colored numbers indicate ranking by Avg. Acceptance Length within each setting: red denotes best, blue denotes second-best, orange denotes third-best.

Setting	Qwen3-32B	Qwen3-4B	Acceptance Length						Stat.
	Stage	Loss	MT-Bench	GSM8K	HumanEval	LongWriter	STEM QA	Avg	Med./P25/P75
Greedy	6 SFT	L_{KL}	4.911	5.827	5.081	4.758	4.709	5.057 (0.005)	6/4/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.915	5.843	5.106	4.792	4.709	5.073 (0.008)	6/4/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	4.956	5.839	5.108	4.805	4.732	5.088 (0.013)	6/4/7
	6 SFT	L_{CE}	4.886	5.803	5.047	4.766	4.680	5.036 (0.007)	6/3/7
	6 SFT	L_{LK}	4.913	5.834	5.092	4.744	4.732	5.063 (0.006)	6/4/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	4.893	5.844	5.067	4.757	4.725	5.057 (0.007)	6/3/7
	6 SFT	L_{WTV}	4.901	5.796	5.066	4.742	4.690	5.039 (0.011)	6/3/7
$T = 0.5$	6 SFT	L_{KL}	4.891	5.770	4.851	4.877	4.485	4.975 (0.007)	6/3/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.914	5.731	4.896	4.893	4.495	4.986 (0.011)	6/3/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	4.981	5.742	4.918	4.892	4.500	5.007 (0.004)	6/4/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	4.929	5.750	4.842	4.879	4.496	4.979 (0.006)	6/3/7
	6 SFT	L_{WTV}	5.394	6.150	5.582	5.296	5.057	5.496 (0.028)	7/4/7
$T = 1.0$	6 SFT	L_{KL}	6.166	6.471	5.318	6.325	5.519	5.960 (0.015)	7/5/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	6.231	6.455	5.401	6.364	5.591	6.008 (0.017)	7/5/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	6.237	6.491	5.421	6.396	5.606	6.030 (0.021)	7/5/7
	6 SFT	L_{CE}	6.590	6.859	6.242	6.840	6.013	6.509 (0.037)	7/7/7
	6 SFT	L_{LK}	6.197	6.463	5.357	6.241	5.554	5.962 (0.016)	7/5/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	6.181	6.480	5.312	6.310	5.519	5.960 (0.019)	7/5/7
	6 SFT	L_{WTV}	6.684	6.895	6.724	6.632	6.422	6.671 (0.005)	7/7/7

L_{WTV} leads by a wide margin, reaching 5.496 at $T = 0.5$ against 4.975 for L_{KL}, and 6.671 at $T = 1.0$ against 5.960.

Table 6 reports the same pair under chain drafting. Under greedy verification the ordering follows the tree case: the GRPO stage on top of $L_{KL} + 0.1L_{EAL}$ is best at 3.636, $L_{KL} + 0.1L_{EAL}$ second at 3.632, and L_{KL} third at 3.615, while L_{WTV} is the weakest objective at 3.603. Under sampling verification L_{WTV} again leads, at 3.640 for $T = 0.5$ and 3.804 for $T = 1.0$ against 3.524 and 3.587 for L_{KL}.

Table 7 reports the acceptance length for Gemma4-31B/Gemma4-E2B under chain drafting. Under greedy verification, L_{EAL} again provides a small but consistent improvement over the L_{KL} baseline, with $L_{KL} + 0.1L_{EAL}$ reaching 5.574 against 5.559. L_{WTV} is competitive but not better in this setting, at 5.558. The advantage of WTV again emerges under sampling verification. At $T = 1$, L_{WTV} achieves 6.102 compared with 5.699 for L_{KL} and 5.722 for $L_{KL} + 0.1L_{EAL}$.

Table 6: Acceptance length on five benchmarks with Qwen3-32B as the target model and Qwen3-4B as the draft model with chain-based speculative decoding under greedy verification and sampling temperatures $T = 0.5$ and $T = 1.0$. Colored numbers indicate ranking by Avg. Acceptance Length within each setting: red denotes best, blue denotes second-best, orange denotes third-best.

Setting	Qwen3-32B	Qwen3-4B	Acceptance Length						Stat.
	Stage	**Loss**	**MT-Bench**	**GSM8K**	**HumanEval**	**LongWriter**	**STEM QA**	**Avg**	**Med./P25/P75**
Greedy	6 SFT	L_{KL}	3.355	4.544	3.412	3.398	3.365	3.615 (0.010)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	3.348	4.536	3.462	3.425	3.387	3.632 (0.005)	3/1/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	3.368	4.501	3.485	3.440	3.385	3.636 (0.008)	3/1/7
	6 SFT	L_{CE}	3.238	4.412	3.369	3.310	3.280	3.522 (0.026)	3/1/7
	6 SFT	L_{LK}	3.351	4.540	3.428	3.367	3.386	3.614 (0.008)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	3.358	4.516	3.442	3.380	3.360	3.611 (0.012)	3/1/7
	6 SFT	L_{WTV}	3.375	4.522	3.425	3.356	3.337	3.603 (0.026)	3/1/7
$T = 0.5$	6 SFT	L_{KL}	3.271	4.438	3.312	3.423	3.174	3.524 (0.002)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	3.270	4.446	3.361	3.475	3.162	3.543 (0.022)	3/1/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	3.313	4.460	3.372	3.505	3.221	3.574 (0.021)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	3.332	4.443	3.297	3.490	3.175	3.547 (0.006)	3/1/7
	6 SFT	L_{WTV}	3.375	4.552	3.420	3.564	3.287	3.640 (0.014)	3/1/7
$T = 1.0$	6 SFT	L_{KL}	3.266	4.420	3.146	3.786	3.316	3.587 (0.025)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{EAL}$	3.290	4.480	3.203	3.858	3.369	3.640 (0.036)	3/1/7
	3 SFT + 3 GRPO	$L_{KL} + 0.1L_{EAL} \rightarrow L_{KL} + 0.1L_{EAL} + 0.1L_{GRPO}$	3.324	4.560	3.167	3.896	3.353	3.660 (0.039)	3/1/7
	6 SFT	L_{CE}	3.242	4.462	3.370	3.920	3.399	3.679 (0.091)	3/1/7
	6 SFT	L_{LK}	3.278	4.448	3.172	3.861	3.343	3.620 (0.029)	3/1/7
	6 SFT	$L_{KL} + 0.1L_{TV}$	3.282	4.526	3.131	3.869	3.366	3.635 (0.025)	3/1/7
	6 SFT	L_{WTV}	3.378	4.599	3.468	4.048	3.526	3.804 (0.005)	3/1/7

Table 7: Acceptance length on three benchmarks with Gemma4-31B as the target model and Gemma4-E2B as the draft model with chain-based speculative decoding under greedy verification and sampling temperature $T = 1.0$. Colored numbers indicate ranking by Avg. Acceptance Length within each setting: red denotes best.

Setting	Gemma4-31B	Gemma4-E2B	Acceptance Length			
	Stage	**Loss**	**MT-Bench**	**GSM8K**	**HumanEval**	**Avg**
Greedy	6 SFT	L_{KL}	4.897	5.984	5.795	5.559
	6 SFT	L_{CE}	4.811	5.873	5.743	5.476
	6 SFT	L_{LK}	4.905	5.996	5.797	5.566
	6 SFT	$L_{KL} + 0.1L_{EAL}$	4.913	6.009	5.800	5.574
	6 SFT	L_{WTV}	4.901	5.977	5.797	5.558
$T = 1.0$	6 SFT	L_{KL}	5.084	6.078	5.936	5.699
	6 SFT	L_{CE}	5.102	6.084	5.957	5.714
	6 SFT	L_{LK}	5.095	6.091	5.944	5.710
	6 SFT	$L_{KL} + 0.1L_{EAL}$	5.108	6.107	5.952	5.722
	6 SFT	L_{WTV}	5.363	6.518	6.424	6.102

5 Conclusion

In this paper we study the design of draft-model training losses that directly target acceptance length in Speculative Decoding. We evaluate different losses across several target–draft pairs, in both chain and tree drafting, and both greedy and sampling-based acceptance criteria. Under greedy verification, adding L_{EAL} on top of the KL anchor gives a consistent improvement over the L_{KL} baseline, and the subsequent GRPO stage adds a further gain. Under sampling verification, L_{WTV} gives a substantially larger improvement at both $T = 0.5$ and $T = 1.0$ in sampling-based speculative decoding. The two losses are therefore complementary, each matching the acceptance criterion it was derived from.

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